

Figure 1 — Defining the drug-indication approval-forecast task

A Prediction-unit comparison

Prior task	Prediction unit	Limitation
LOA analysis	disease / phase / modality rate	not individualized
Drug-level approval	drug	not disease-specific
Trial-outcome prediction	trial	not final approval, often trial-feature dependent

This work drug-indication pair → individualized, disease-specific, approval-level (trial-agnostic)

B Dataset construction pipeline



C Final cohort summary

14,134
drug-indication pairs

2,469 / 11,665
approved / failed

2,637
test pairs (624 positive)

2,313
unique drugs

4,190
unique indications

5.18
avg indications / drug